

Lab 02

Object Oriented Programming

BS DS/CY Fall 2025 Morning/Afternoon

Instructions:

- Indent your code properly.
- Use meaningful variable names and follow standard naming conventions.
- Use clear and meaningful prompts/labels for all program input and output.
- Do not discuss your problems with fellow students. If you face any difficulty in understanding, consult your TAs.

Task 01

Implement following function named **getPosNeg** that accept an array **ar** along with its size **n_ar**

```
void getPosNeg(const int ar[], const int n_ar, int* &pos, int& n_pos, int* &neg, int& n_neg);
```

The parameters **ar** and **n_ar** hold an array and its size respectively.

The function gets all the positive and negative numbers from the array **ar** and places them into newly created arrays pointed by parameter **pos** and **neg** respectively. Store the sizes of **pos** and **neg** arrays into **n_pos** and **n_neg** respectively. It should store 0 (zero) and **nullptr** in parameters **n_pos/n_neg** and **pos/neg** respectively, if **ar** has no positive and or negative numbers. *The function should not display anything.*

In the **main** function declare an array of size 10. Fill the array with arbitrary values and then pass it to **getPosNeg** function along with its size and all the required parameters. After the execution of function display contents of the arrays pointed by **pos** and **neg**. Display appropriate message(s), if **pos** and or **neg** arrays are empty. Don't forget to free the memory resource allocated by the program, if any.

Task 02

<pre>struct Book { char name[50]; char publisher[50]; int ISBNNumber; };</pre>	<pre>struct Library { Book * data; int noOfBooks; int capacity; };</pre>
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Struct Description:

Book:

It will store the item information which user wants to buy.

- 'name' attribute will store book name
- 'publisher' will store name of the publisher.
- 'ISBNNumber' will store the ISBN number of book.

Library:

It will store/contain all available books.

- 'data' attribute will point to an array of 'Book' on heap, which will contain the book available at library.
- 'noOfBooks' represents the total no of Books available in library.
- 'capacity' represents the size of library i.e. the size of array pointed by 'data'.

Functions:

- void initializeLibrary(Library & ly, const int capacity);
- bool addBook(Library & ly, const Book & bo);
- bool removeBook (Library& ly, char * BookName, int count = 1);
- void DisplayAllBooksInformation(Library & ly);
- void resize(Library& ly , int s=5);

Task 03

Creating a Shopping Cart and Point of Sale System.

<pre>struct Item { char name[50]; int quantity; float unitPrice; };</pre>	<pre>struct Cart { Item * data; int noOfItems; int capacity; };</pre>
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Struct Description:

Item struct:

It will store the item information which user wants to buy.

- 'name' attribute will store item name
- 'quantity' will *store number of items to buy*
- 'unitPrice' will store price per item.

Cart struct:

It will store/contain the items which user wants to buy and put in his shopping basket.

- 'data' attribute will point to an array of 'Item' on heap, which will contain the items bought by user.
- 'noOfItems' *represents the no of items bought by user.*
- 'capacity' represents the size of cart i.e. the size of array pointed by 'data'.

Functions:

- void initializeCart(Cart & c, int capacity);
- bool addItem(Cart & c, const Item & it);
- bool removeItem(Cart & c, char * itemName, int count = 1);
- void checkOut(Cart & c);
- void resize(int s=5);